Rishi Upath

Electrical Engineering Student | Hardware Design & Validation | Vancouver, BC

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Education

University of British Columbia – BASc in Electrical Engineering (Year 3, GPA: 87%) Expected May 2028
Co-op Program: Available 4–8 months beginning May 2027

Experience

Hardware Validation Co-op, Microchip Technology – Burnaby, BC September 2026 – Present

- Performing post-silicon validation of high-performance spaceflight computing processor (PIC64-HPSC) through hardware platform testing and board-level firmware development in Linux environment using C++ and Python
- Developed firmware feature enabling runtime configuration of onboard jitter attenuator from a resource constrained MCU running FreeRTOS, porting C++ I2C driver, creating Python API, and validating functionality using oscilloscope
- Debugging complex 20+ layer evaluation boards using Altium Designer schematics/layouts, scripting, and lab equipment

Hardware Engineer Co-op, NETGEAR – Richmond, BC January 2026 – August 2026

- Developed and executed design verification test plans for mobile hotspot hardware using oscilloscopes, power analyzers, and temperature chambers, identifying 30+ bugs and collaborating across teams to root-cause failures and verify fixes
- Prototyped low-cost ESP32-driven LCD demonstration system for in-store display units, rapidly fabricating adapter boards via Altium and in-house PCB plotter, potentially saving $\$800,000$ vs. fully built units
- Interpreted schematics and layouts in Cadence Allegro to perform 0201 SMT rework and board modifications for debug

Electrical Team Member, UBC Rover – Vancouver, BC September 2024 – Present

- Designed rover electrical subsystems, including power distribution, remote stop, and wiring system under tight deadlines
- Authored safety compliance documentation and completed functional safety testing of entire rover electrical system, successfully passing judge safety inspection at the 2026 Canadian International Rover Challenge (CIRC)

Projects

Rover Power Distribution System (UBC Rover) – Altium, Power Electronics, STM32, CAN FD, UART Portfolio

- Developed, assembled, and deployed a 3-board power distribution system in Altium Designer, generating 24 V, 18 V, 12 V, and 5 V rails from a 7S3P LiPo battery, successfully powering the rover throughout CIRC 2026 with zero power faults
- Designed 20 A synchronous buck converters and a 100 A-rated 4-layer PCB, sizing high-current copper per IPC-2221 and coordinating fuse, connector, conductor, and thermal ratings to meet competition safety requirements
- Implemented rail voltage and current monitoring using Hall-effect current sensors and STM32 ADC sampling, developing firmware for inter-board UART and CAN FD telemetry with software subsystem

Vehicle Remote Stop (UBC Rover) – Altium, LTspice, Relays, LoRa Radio, I2C, CAN FD, UART, USB Portfolio

- Leading end-to-end development of 3 custom PCBs for long-range emergency remote stop system using LoRa radio and relay-based power disconnect, including simulation, component selection, schematics, and layout in Altium and LTspice
- Implementing STM32-based hardware architecture, integrating I2C, UART, and CAN FD communication interfaces, switching/LDO regulators, and design-for-debug including 0 Ohm jumpers and power rail test points

USB Cable Current Monitor (NETGEAR) – Altium, PCB Manufacturing Portfolio

- Designed and manufactured 4 USB cable current/voltage monitor PCBs in Altium Designer, enabling more accurate and reliable USB-C charging validation of consumer electronic devices
- Independently fabricated PCBs in-house using LPKF PCB plotter, eliminating external manufacturer lead times and cost
- Implemented Kelvin 4-wire sensing with large copper pours to minimize measurement error, added RC low-pass filter to reject power supply noise, and characterized sense resistor via V–I curve fitting to correct for tolerance error

FPGA ARC4 Decryption Algorithm – Quartus, ModelSim, SystemVerilog, FPGA December 2025

- Designed ARC4 decryption engine in SystemVerilog on DE1-SoC FPGA, using hierarchical architecture and on-chip RAM
- Achieved 90x speedup over single core performance by instantiating 72 parallel cracking modules, increasing clock frequency using PLL, and analyzing worst-case slack paths in Quartus Timing Analyzer
- Developed comprehensive ModelSim testbenches for both RTL and post-synthesis simulations to verify functionality

Skills

Software: Altium Designer, Cadence Allegro, LTspice, Git, Quartus, ModelSim, Jira
Programming: C/C++, Python, Linux, FreeRTOS, SystemVerilog, PlantUML, Assembly
Lab Equipment: Multimeter, Oscilloscope, Function Generator, Power Analyzer, Soldering, Temperature Chamber, PCB Fab